



Sreeja T 251801325-2025-AIDS ▾

S2

Started on	Thursday, 6 August 2026, 10:11 AM
State	Finished
Completed on	Thursday, 6 August 2026, 10:25 AM
Time taken	14 mins 5 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
```

```
{
```

```
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
```

```
    {
```

```
        i++;
```

```
        s += i;
```

```
    }
```

```
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```

1  #include<stdio.h>
2  int main(){
3      int n;
4      int i=1;
5      int s=1;
6      int count = 2;
7      scanf("%d",&n);
8      while(s<=n){
9          count++;
10         i++;
11         count++;
12         s+=i;
13         count++;
14     }
15     count++;
16     printf("%d",count);
17     return 0;
18 }
```

	Input	Expected	Got	
✓	9	12	12	✓
✓	4	9	9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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S2**Started on** Thursday, 6 August 2026, 10:27 AM**State** Finished**Completed on** Thursday, 6 August 2026, 10:42 AM**Time taken** 15 mins 13 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
        printf("*");
    }
    else
    {
        for(int i=1; i<=n; i++)
        {
            for(int j=1; j<=n; j++)
            {
                printf("*");
                printf("*");
                break;
            }
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```
1  #include<stdio.h>
2  int main() {
3      int n;
4      scanf("%d",&n);
5      int count=0;
6      count++;
7      if(n==1){
8          count++;
9          //printf("*");
10     }
11     else{
12         for(int i=1;i<=n;i++){
13             count++;
14             for(int j=1;j<=n;j++){
15                 count++;
16                 //printf("*");
17                 count++;
18                 //printf("*");
19                 count++;
20                 break;
21             }
22             count++;
23         }
24         count++;
25     }
26     printf("%d",count);
27     return 0;
28 }
```

	Input	Expected	Got	
✓	2	12	12	✓
✓	1000	5002	5002	✓
✓	143	717	717	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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S2**Started on** Thursday, 6 August 2026, 10:46 AM**State** Finished**Completed on** Thursday, 6 August 2026, 10:50 AM**Time taken** 4 mins 9 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
    for (i = 1; i <= num; ++i)
    {
        if (num % i == 0)
        {
            printf("%d ", i);
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1 #include<stdio.h>
2 int main(){
3     int n;
4     scanf("%d",&n);
5     int count=0;
6     for(int i=1;i<=n;i++){
7         count++;
8         count++;
9         if(n%i==0){
10            count++;
11            //printf("%d",i);
12        }
13    }
14    count++;
15    printf("%d",count);
16    return 0;
17 }
```

	Input	Expected	Got	
✓	12	31	31	✓
✓	25	54	54	✓
✓	4	12	12	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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S2

Started on	Thursday, 6 August 2026, 10:50 AM
State	Finished
Completed on	Thursday, 6 August 2026, 10:57 AM
Time taken	6 mins 33 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c = 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include<stdio.h>
2  int main(){
3      int n;
4      scanf("%d",&n);
5      int c=0;
6      int count=1;
7      for(int i=n/2;i<n;i++){
8          count++;
9          for(int j=1;j<n;j=2*j){
10             count++;
11             for(int k=1;k<n;k=k*2){
12                 count++;
13                 c++;
14                 count++;
15             }
16             count++;
17         }
18         count++;
19     }
20     count++;
21     printf("%d",count);
22     return 0;
23 }
```

	Input	Expected	Got	
✓	4	30	30	✓
✓	10	212	212	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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S2**Started on** Thursday, 6 August 2026, 10:57 AM**State** Finished**Completed on** Thursday, 6 August 2026, 11:06 AM**Time taken** 8 mins 46 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
    int rev = 0, remainder;
    while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;
    }
    print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include<stdio.h>
2  int main(){
3      int n;
4      scanf("%d",&n);
5      int rev=0,remainder;
6      int count=1;
7      count++;
8      while(n!=0){
9          count++;
10         remainder = n%10;
11         count++;
12         rev=rev*10+remainder;
13         count++;
14         n/=10;
15         count++;
16     }
17     //printf(rev);
18     count++;
19     printf("%d",count);
20     return 0;
21 }
```

	Input	Expected	Got	
✓	12	11	11	✓
✓	1234	19	19	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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